Explainable Artificial Intelligent (XAI)

MAKINI CALEB

2026-10-08

SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

load libraries

```
library(randomForest)
library(DALEX)
library(ggplot2)
library(xgboost)
library(rio)
library(caret)
library(ggplot2)
library(dplyr)
library(Matrix)
library(Metrics)
```

import inbuilt data

```
data(mtcars)
mtcars$am <- factor(mtcars$am, levels = c(0, 1), labels = c("Automati
c", "Manual"))
df <- mtcars[, c("am", "mpg", "hp", "wt", "cyl")]
head(df)
```

```
                       am  mpg  hp    wt cyl
Mazda RX4            Manual 21.0 110 2.620   6
Mazda RX4 Wag        Manual 21.0 110 2.875   6
Datsun 710           Manual 22.8  93 2.320   4
Hornet 4 Drive    Automatic 21.4 110 3.215   6
Hornet Sportabout Automatic 18.7 175 3.440   8
Valiant           Automatic 18.1 105 3.460   6
```

train random forest model

```
set.seed(42)
rf_model <- randomForest(am ~ ., data = df, ntree = 100, importance =
TRUE)
print(rf_model)
```

```

Call:
 randomForest(formula = am ~ ., data = df, ntree = 100, importance = T
RUE)
               Type of random forest: classification
                     Number of trees: 100
No. of variables tried at each split: 2

        OOB estimate of  error rate: 12.5%
Confusion matrix:
          Automatic Manual class.error
Automatic        18      1  0.05263158
Manual            3     10  0.23076923
```

create a DAlex explianer

- the explainer will act as a bridge btn the machine learning model and xai tools

```
y_num <- ifelse(df$am == "Manual", 1, 0)
explainer <- DALEX::explain(model = rf_model, data = df[, -1], y = y_n
um, label = "Random Forest",
    verbose = FALSE)
```

global excplainability? feature importance

which model influence the model the most?

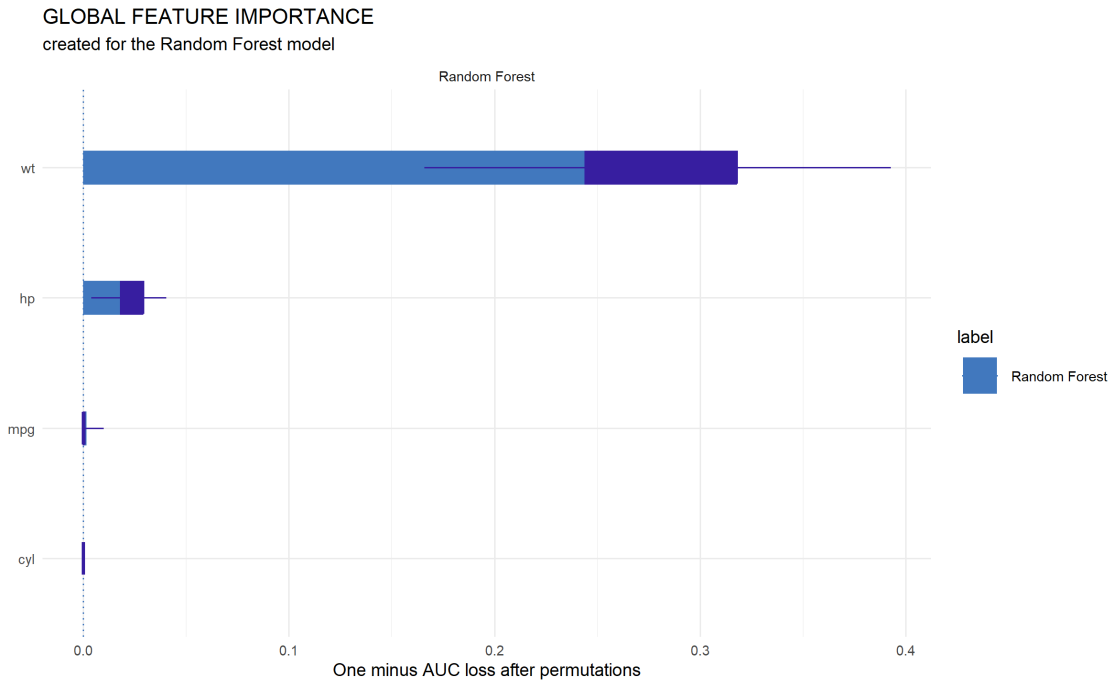
- Dalex measures feature importance by shuffling variables and observing how much model perform

```
global_importance <- model_parts(explainer, type = "variable_importanc
e")
global_importance
```

```
      variable mean_dropout_loss         label
1 _full_model_       0.000000000 Random Forest
2          cyl       0.000000000 Random Forest
3          mpg       0.001619433 Random Forest
4           hp       0.023684211 Random Forest
5           wt       0.289878543 Random Forest
6   _baseline_       0.518623482 Random Forest
```

plot global importance

```
plot(global_importance) + ggtitle("GLOBAL FEATURE IMPORTANCE") + theme
_minimal()
```



interpretation - the lager the more important the variable is

set a native Random fOrest Importance

```
importance(rf_model)
```

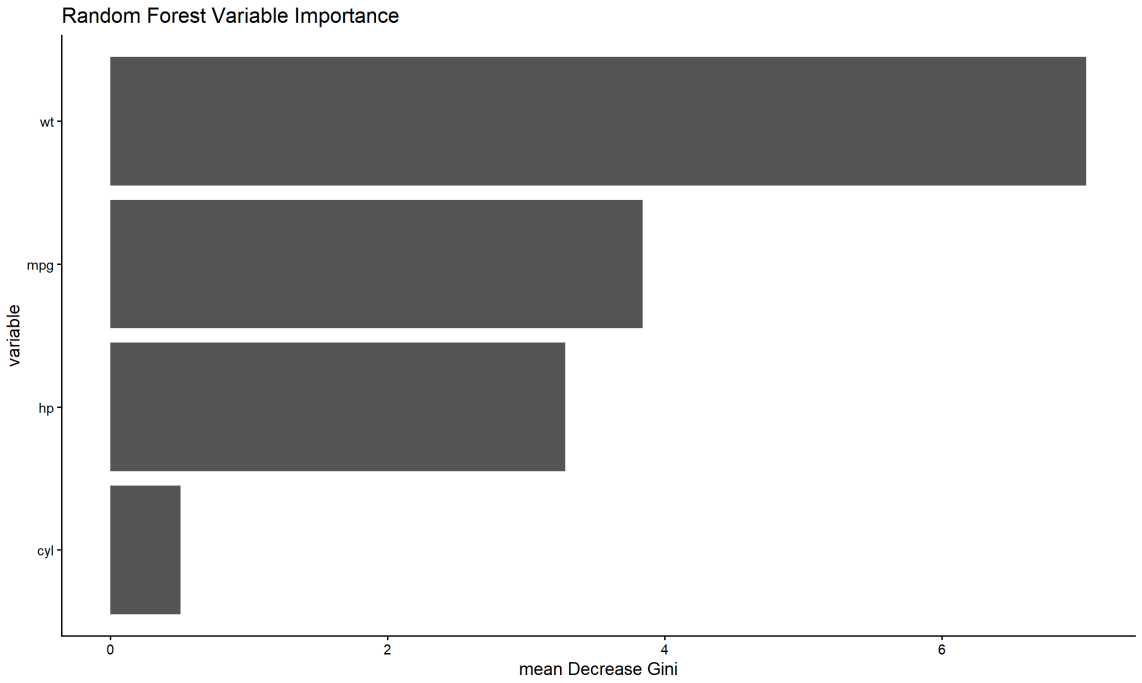
```
     Automatic     Manual MeanDecreaseAccuracy MeanDecreaseGini
mpg  2.4483701  5.0830670           4.39618934        3.8419660
hp   0.8357003  2.3170812           2.15807215        3.2832931
wt  11.0436722 11.1837422          13.58513706        7.0424824
cyl  0.2159529  0.7676641           0.04610337        0.5058002
```

convert into a bar chart

```
importance_df <- data.frame(Variable = rownames(importance(rf_model)),
Importance = importance(rf_model)[,
    "MeanDecreaseGini"])
importance_df <- importance_df[order(importance_df$Importance, decreas
ing = TRUE),
    ]
```

plotting

```
ggplot(importance_df, aes(x = reorder(Variable, Importance), y = Impor
tance)) + geom_bar(stat = "identity") +
    coord_flip() + labs(title = "Random Forest Variable Importance ",
x = "variable",
    y = "mean Decrease Gini") + theme_classic()
```



select car

```
new_car <- df[1, -1]
new_car
```

```
          mpg  hp   wt cyl
Mazda RX4  21 110 2.62   6
```

make prediction

```
predict(rf_model, new_car)
```

```
Mazda RX4
   Manual
Levels: Automatic Manual
```

actual class

```
df$am[1]
```

```
[1] Manual
Levels: Automatic Manual
```

generate local explanation

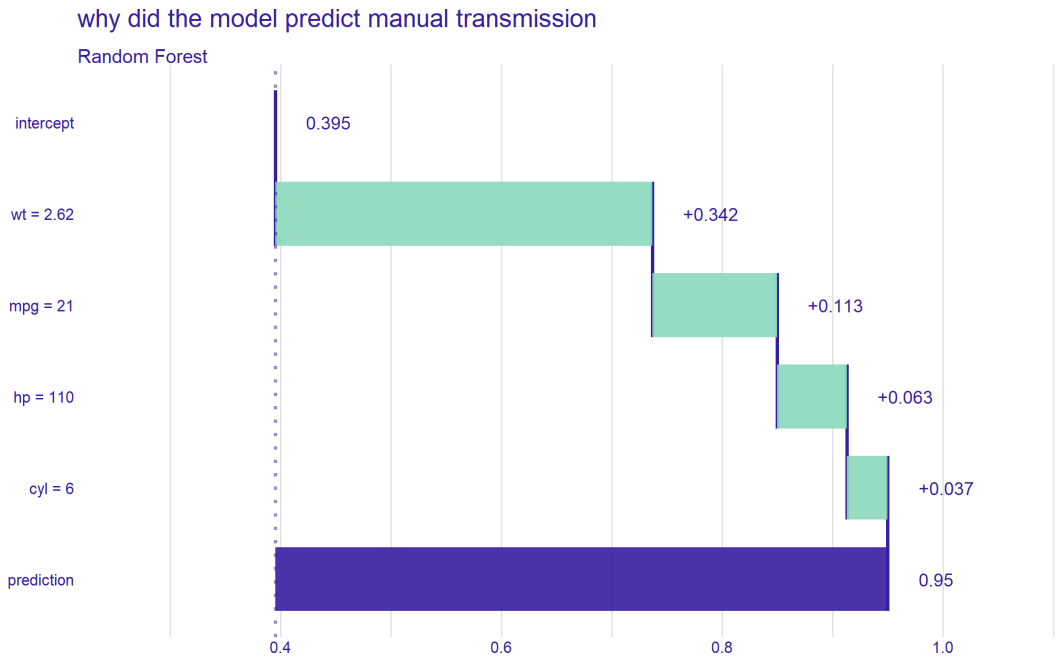
- breakdown explanation show how each variable contribute to global to a single prediction

```
local_explanation <- predict_parts(explainer, new_observation = new_ca
r, type = "break_down")
print(local_explanation)
```

```
                          contribution
Random Forest: intercept         0.395
Random Forest: wt = 2.62         0.342
Random Forest: mpg = 21          0.113
Random Forest: hp = 110          0.063
Random Forest: cyl = 6           0.037
Random Forest: prediction        0.950
```

plot local explanation

```
plot(local_explanation) + ggtitle("why did the model predict manual tr
ansmission")
```


why did the model predict manual transmission
Random Forest
intercept
wt = 2.62
mpg = 21
hp = 110
cyl = 6
prediction
0.395
+0.342
+0.113
+0.063
+0.037
0.95
0.4
0.6
0.8
1.0